

Chapter 3

Methodology

This chapter details the tools and methods used to both generate the improved DNP group parameters and to validate them. Three models are covered in this chapter: the 0D scaled model, which is already complete; the 0D flow model, which is proposed; and the spatially resolved model, which is discussed for potential avenues of future research. Although each model varies in its fidelity and cost, the general methodology is very similar: generate concentrations of DNPs from irradiation of a sample, convert those concentrations into a delayed neutron count rate, and then use non-linear least squares to fit DNP group parameters to the count rate curve. Validation of the DNP group parameters requires solving kinetics problems that make use of the parameters. There are also transient problems of interest that affect safety, security, and safeguards by design.

3.1 Tools

Several external software tools evaluate and generate the improved DNP group parameters. Molten Salt Delayed Neutron (MoSDeN), an in-house developed Python package, collects and combines data, OpenMC is used for data handling and higher fidelity analysis, and the MOOSE app Moltres evaluates and refines them at different fidelity levels [183, 149, 105]. Preliminary results use MoSDeN to generate the improved DNP group parameters, while the proposed work will use OpenMC and Moltres. Each tool is described in detail in its corresponding sub-section.

3.1.1 MoSDeN

MoSDeN is a Python package designed to enable generation of DNP group parameters using the microscopic approach for any given reactor of interest. MoSDeN allows for varying levels of fidelity in the generation, ranging from only using nuclear data and broad assumptions to a high-fidelity multiphysics simulation with neutronics and thermal-hydraulics. The MoSDeN package relies on the SciPy, NumPy, Matplotlib, pandas, and Uncertainties packages [187, 70, 77, 179, 55]. Many features of MoSDeN also require OpenMC to function properly [149]. MoSDeN automates data collection and processing, simulates pulse and/or saturation irradiations, converts the concentrations into a delayed neutron count rate, and generates group parameters.

The preliminary work on MoSDeN uses the 0D scaled model, while the proposed work will make use of the 0D flow model. Both models are discussed in more detail in Section 3.2, as is the spatially resolved model, which is proposed for future work.

3.1.2 OpenMC

OpenMC is an open-source Monte Carlo neutral particle transport code. Its Python Application Programming Interface (API) offers functionality that simplifies interfacing with MoSDeN, an in-house developed open-source Python package. The sample irradiation process is straightforward and configurable, as users can adjust parameters through an input file. OpenMC also offers useful feature in depletion, where a zero-power depletion step is rapidly calculated since only decay is tracked. This feature allows resolving ex-core residence time depletion steps quickly. Additionally, OpenMC's transfer rate functionality during depletion enables simulation of chemical removal in an MSR.

For this work, MoSDeN by default uses the ENDF/B-VII.1 based depletion chains provided by OpenMC and ENDF/B-VII.1 cross section data. The thermal spectrum chain applies to neutron irradiation below 1 *eV*, the fast spectrum chain applies to neutron irradiation above thermal and below 5 *MeV*, and anything above that is not treated in this work [172]. The models by default assume monoenergetic neutrons irradiate the sample.

3.1.3 Moltres

Moltres is an open-source MOOSE application that simulates MSRs by solving the multi-group neutron diffusion and Navier-Stokes equations [105]. MOOSE is an open-source, parallel, finite element framework for multiphysics applications [66]. MOOSE includes many pre-built applications that couple physical phenomena. The HEAT CONDUCTION and NAVIER-STOKES modules are particularly useful in this work, as they can couple with Moltres for general thermal-hydraulic modeling [131, 104, 133, 129]. Moltres will simulate the MSRE with different DNP group parameters. Moltres will also validate the improved DNP group parameters in this work.

The MSRE reactivity data can be used for validation, and was previously modeled with Moltres [105]. The MSRE core consists of flow through channels, simplifying its modeling compared to open-pool designs due to the restricted flow path. Since the initial demonstration, Park et al. have noted general agreement in results, attributing differences to simplifications and assumptions in the geometry [129]. Additionally, Park has published a dissertation detailing 1-D, 2-D, and 3-D neutronic models of the MSRE [128].

Moltres requires various data to model various physical phenomena in a reactor. The delayed neutron modeling requires delayed neutron fractions, β_k , and delayed neutron decay constants, λ_k , which are fission and adjoint-weighted. Neutronics modeling requires cross-sections, prompt and delayed fission spectra, and other data to properly model those physical effects. Moltres provides scripts which can parse various transport solvers, such as OpenMC and Serpent, to automatically collect this information and format it for use in Moltres.

There are two main capabilities in Moltres which are of interest. Moltres couples three physical phenomena to model an MSR: neutron flux, DNPs, and thermal-hydraulics. This allows Moltres to model transient simulations of MSRs, which will be used to validate the improved DNP group parameters. Moltres can also simulate transients using the MSR-specific DNP group parameters, and it can generate improved parameters with higher fidelity than the other methods. This is of interest for future work, which may use higher-fidelity multi-physics to generate improved DNP group parameters.

Neutron Flux Modeling

Neutron flux modeling determines the energy of neutrons interacting with the sample and the number of interactions. It determines the fission rate and the resulting fission product concentrations, including DNPs. The time-dependent multi-group neutron diffusion equation appears in Equation (2.34), recreated here:

$$\begin{aligned}
\frac{1}{v_g} \frac{\partial \phi_g}{\partial t} = & \nabla \cdot D_g \nabla \phi_g + \sum_{g' \neq g}^G \Sigma_{s,g' \rightarrow g} \phi_{g'} \\
& + (1 - \beta) \chi_{p,g} \sum_{g'=1}^G (v \Sigma_f)_{g'} \phi_{g'} \\
& - \sum_{i=1}^I \chi_{d,g,i} \lambda_i C_i - \Sigma_{a,g} \phi_g - \sum_{g' \neq g}^G \Sigma_{s,g \rightarrow g'} \phi_g
\end{aligned}$$

Modeling the neutron flux requires accounting for the DNPs. DNPs contribute to the neutron flux and possess their own energy spectra, necessitating their inclusion as a separate term.

DNP Modeling

The time dependent DNP concentration is given in Equation (2.35) and repeated here:

$$\frac{\partial C_i}{\partial t} = \beta_i \sum_{g=1}^G (v \Sigma_f)_g \phi_g - \lambda_i C_i - \nabla \cdot (C_i \vec{u}) + \nabla \cdot \frac{v_T}{S c_T} \nabla C_i$$

This is the primary equation of interest when validating the improved DNP group parameters.

For the future work involving formulation of improved DNP group parameters within Moltres for the spatially resolved model, this equation is less helpful. This is because it requires DNP group parameters, which Moltres should be generating. To generate the DNP group parameters, this equation is replaced by (2.58), recreated here:

$$\frac{dN_j}{dt} = S(\vec{r}, t) - \lambda_{net,j} N_j(\vec{r}, t) + \nabla \cdot (D \nabla N_j(\vec{r}, t) - \vec{u}(\vec{r}, t) N_j(\vec{r}, t))$$

The spatially resolved model will use spatially resolved depletion over the irradiation simulation. The change to the equation results in tracking concentrations of the individual DNPs instead of tracking concentrations of DNP groups.

Thermal-Hydraulic Modeling

Moltres offers time-dependent, incompressible, buoyancy-driven flow [128]:

$$\rho \frac{\partial \vec{u}}{\partial t} = -\rho(\vec{u} \cdot \nabla) \vec{u} - \nabla p + \mu \nabla^2 \vec{u} + \rho \alpha \vec{g} (T - T_{\text{ref}}), \quad (3.1)$$

$$\nabla \cdot \vec{u} = 0, \quad (3.2)$$

where

$$\begin{aligned}
\rho &= \text{fluid density } [kg \cdot m^{-3}] \\
p &= \text{pressure } [Pa] \\
\vec{u} &= \text{flow rate } [m \cdot s^{-1}] \\
\mu &= \text{dynamic viscosity } [Pa \cdot s] \\
\alpha &= \text{coefficient of thermal expansion } [K^{-1}] \\
\vec{g} &= \text{gravitational force vector } [N] \\
T &= \text{fluid temperature } [K] \\
T_{\text{ref}} &= \text{reference temperature at which the nominal} \\
&\quad \text{density is provided } [K].
\end{aligned}$$

The temperature distribution is given by [128] as:

$$\rho c_p \frac{\partial T}{\partial t} = -\rho c_p \vec{u} \cdot \nabla T + \nabla \cdot (k \nabla T) + Q_f - Q_s, \quad (3.3)$$

where the fission heat source term is defined [128] as:

$$Q_f = \sum_{g=1}^G \epsilon_g \Sigma_g^f \phi_g, \quad (3.4)$$

where

$$\begin{aligned}
c_p &= \text{specific heat capacity of molten salt } [J \cdot kg^{-1} \cdot K^{-1}] \\
k &= \text{effective thermal conductivity of molten salt } [W \cdot m^{-1} \cdot K^{-1}] \\
Q_f &= \text{fission heat source } [W \cdot m^{-3}] \\
\epsilon_g &= \text{average fission energy released by neutrons in group } g [J] \\
Q_s &= \text{heat sink/removal } [W \cdot m^{-3}].
\end{aligned}$$

With the combination of the flux modeling, precursor modeling, and thermal-hydraulic modeling, Moltres can almost simulate MSRs. However, there is one additional MSR phenomenon required to complete the model, the flow of the fuel salt between the in-core to ex-core regions.

Ex-core and In-core Coupling

Moltres treats the ex-core region as a tightly-coupled 1D pipe geometry with coupling using Picard iterations and the inlet/outlet boundary conditions. The specific equations to treat the variables of interest in this geometry can be derived from their full forms. For example, the flux and eddy viscosity are zero in the ex-core region, leading to the precursor concentration in the ex-core region:

$$\frac{\partial C_i}{\partial t} = -\lambda_i C_i - u_x \frac{\partial C_i}{\partial x}. \quad (3.5)$$

3.2 DNP Group Generation

Generating DNP group parameter data in a reactor-agnostic manner requires a reactor-agnostic generation process. Fully detailing the flow regime with a large number of parameters is computationally infeasible. Instead, the flow is simplified to two parameters: an in-core average residence time, τ_{in} , and an ex-core residence time, τ_{ex} .

This approach captures fuel salt advection around the primary loop and approximates turbulent and diffusive mixing. The terms derive from the averaged behavior of a fuel salt control volume, treating thermal-hydraulic effects, such as vortices, as changes in residence times. This method allows for approximate fitting of any circulating fuel reactor, requiring only the generation of relevant parameters for determining the associated DNP group parameters.

The other required parameters describe chemical removal, which varies by fission product and reprocessing rate. Refueling could also be of interest, but only a single sample of fissile material is considered during irradiation. The introduction of fresh fuel would affect the incident neutron energy spectrum on the fissile sample, and is considered in Section 5.4. An irradiation time of approximately seven minutes suffices to reach equilibrium DNPs concentrations, as the saturation time frame is on the order of multiple half-lives of the longest-lived DNP. Elements with cycle times on the order of months to years exhibit negligible changes, as discussed in Section 4.2.1. Elements with changes on the order of seconds to days contribute sufficiently and must be included for accurate data generation.

The MSBR reprocessing scheme serves as the basis for analyzing the most extreme cycle times in this work. This is because the MSBR presents very large chemical removal rates, providing useful conservative estimates as to the impact these large removal rates could have on DNP group parameters. These chemical removal rates are given in Table 4.3. Rare earths, semi-noble metals, volatile fluorides, discard, and higher nuclides have small removal rates relative to the time scales of interest for saturation irradiations. Section 4.2.1 examines these longer cycle times, the inverse of the removal rates, to confirm their negligible impact. Section 4.2.1 scales the cycle times to analyze how reprocessing rates influence the group parameters.

Table 3.1: MSBR Online Reprocessing Removal Rates from [146].

Reprocessing Group	Element(s)	Removal Rate [s^{-1}]
Rare Earths	Y, La, Ce, Pr, Nd, Pm, Sm, Gd	2.32×10^{-7}
Rare Earths	Eu	2.32×10^{-8}
Noble Metals	Se, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Sb, Te	5.00×10^{-2}
Seminoble Metals	Zr, Cd, In, Sn	5.79×10^{-8}
Gases	Kr, Xe	5.00×10^{-2}
Volatile Fluorides	Br, I	1.93×10^{-7}
Discard	Rb, Sr, Cs, Ba	3.37×10^{-9}
Salt Discard	Th, Li, Be, F	3.37×10^{-9}
Protactinium	Pa	3.86×10^{-6}
Higher Nuclides	Np, Pu	1.98×10^{-9}

With the chemical reprocessing scheme in place, the next step to generate the improved DNP group parameters is to decide on the level of fidelity. The three levels of fidelity are the 0D scaled model, the 0D flow model, and the spatially resolved model. The 0D scaled model is completed, the 0D flow model is planned, and the spatially resolved model is a potential future development.

A Monte Carlo sampling approach is used to propagate uncertainties in the various inputs, such as in emission probabilities, half-lives, fission yields, reprocessing terms, or residence times. The sampling approach simply repeats the process for generating the DNP group yields, but with randomly sampled input data. The data itself must be perturbed, and not the delayed neutron count rate itself. This is because the uncertainties in the delayed neutron count rate are not independent from time step to time step, but instead they all depend on the underlying data which changes the shape of the delayed neutron count rate at all times.

For generation of the improved DNP group parameters, the main goal is to formulate them for individual fissile nuclides. For some of the analyses presented in this work, the fissions do not come from a single fissile nuclide. For these analyses, a fission-weighted approach is used, where the DNP group parameters are scaled based on the fraction of fissions that lead to those group parameters. Equations (2.13) and (2.14) show how the fission-weighted DNP group parameters can be calculated. The fission-weighted values can then be multiplied by an importance term to account for the difference between the delayed and prompt neutron fission spectra, as shown in Equations (2.15) and (2.16) [141].

3.2.1 0D Scaled Model

This is the lowest fidelity of the three models, and it primarily serves as a way to verify the microscopic approach is working as expected by comparing with existing literature. This approach can only model saturation concentrations, as there is no decay chain tracking nuclides decaying into each other. It is called the 0D scaled model because it uses the scaled flux method, discussed in Section 2.2.2. The cumulative fission yield is used to calculate the equilibrium DNP concentrations, with a scaling term based on the amount of fuel salt in the core. An example solution with no reprocessing is shown in Equation (3.6) as [79]:

$$N_i = \frac{f_{in} C F Y_i}{\lambda_i}, \quad (3.6)$$

where

f_{in} = the fraction of fuel-salt in the core.

The derivation for this equation is given in Appendix B. It is important to note that the flux is scaled when calculating the fission rate as well, by the same f_{in} term that scales the cumulative fission yield. Effectively, this cancels the term out, resulting in no deviation in the results based on the in-core fuel salt fraction by reprocessing time.

No parasitic absorption is accounted for in this model, nor is time dependent decay of nuclides into DNPs. There is a possible approach to include the effects of parasitic absorption, but this approach is not included [164, Benedict et al.].

Figure 3.1 shows the how the different data is loaded into MoSDeN, including external data and user defined input data. The green circles represent input data required for generating the DNP group parameters. The red diamond represents MoSDeN, the tool takes in the data to simulate irradiation of the target fissile sample to generate the irradiated concentrations, the delayed neutron count rate, and finally the DNP group parameters that fit that count rate. The blue squares represent the outputs, in this case the DNP group parameters: the DNP group yields

and the DNP group half-lives.

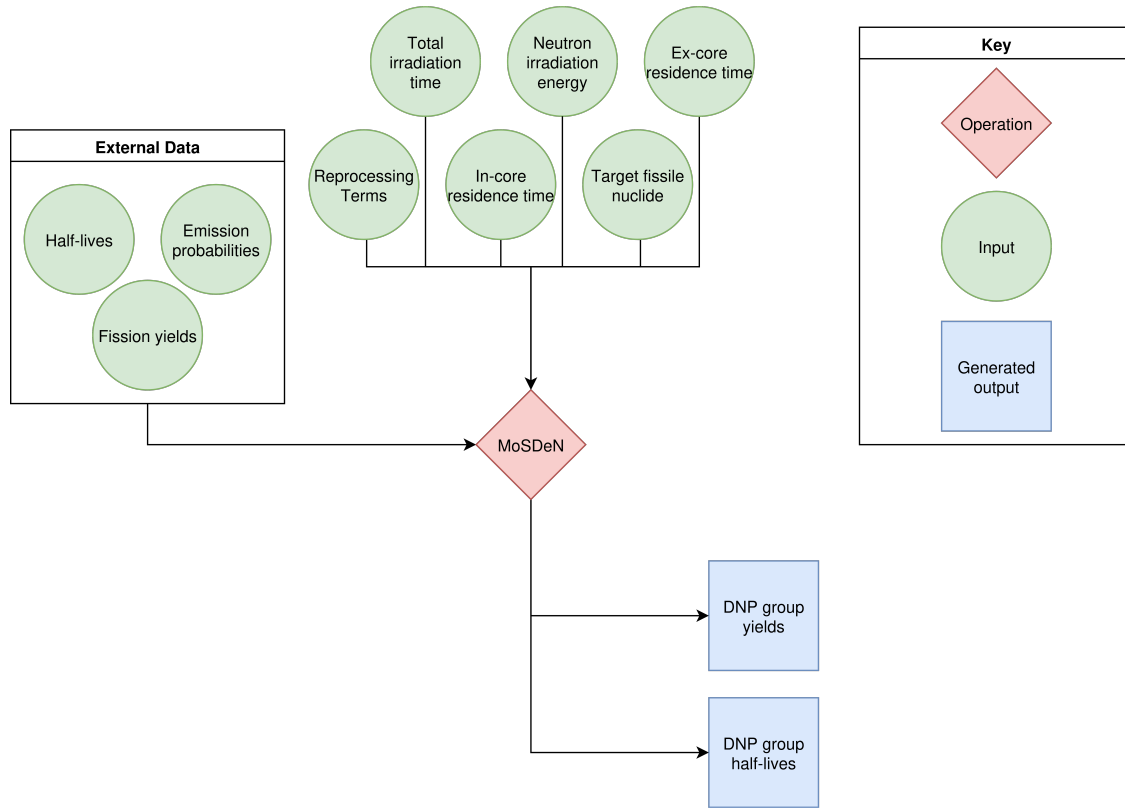


Figure 3.1: Flow diagram of 0D scaled model used to generate improved DNP group parameters.

After all this data is loaded into MoSDeN, it generates the concentration of each DNP. This concentration is then combined with emission probabilities and decay constants to form a delayed neutron count rate. A non-linear least squares solve then fits the improved DNP group parameters to the delayed neutron count rate, generating the group yields and half-lives, given as blue squares in Figure 3.1.

This model is only valid for saturation irradiation simulations, as the concentration calculated in Equation (3.6) is the equilibrium concentration. For a pulse irradiation, this approach is not valid. There are four different options available to handle pulse irradiations: reduce net decay measurement time, pull only the shortest lived group(s) with independent fission yields, use OpenMC depletion, or simply do not use the shortest lived groups typically generated from a pulse irradiation since they are unlikely to be changed by MSR physical phenomena. The first approach, reducing the net decay time, will result in overfitting of the short lived groups, leading to less accurate results. The second approach, using only the shortest lived groups with independent fission yields means that the decay chains will not be handled, but the initial pulse concentration should be accurate. However, the concentrations will quickly become less accurate as the decays of various nuclides into DNPs will not be treated, so this approach could only be potentially viable for the shortest lived group. The third approach, using OpenMC depletion, would be accurate and easily achievable. However, it would greatly increase computational cost, and the use of OpenMC in such a case is already handled by the 0D flow model, making it redundant. The final approach would entail only using the longer lived DNP group parameters from a saturation irradiation, while recognizing that the short lived groups are less accurate. Because the MSR physical phenomena that cause non-negligible influence on the DNPs take at least several seconds to come into effect, it is reasonable to focus on the longer lived groups.

3.2.2 0D Flow Model

The 0D flow model enables improved analysis of the general trends in nuclide evolutions. It treats the flow in a zero-dimensional manner using in-core and ex-core residence times, and uses OpenMC to track the full depletion matrix. The fission history function is a square wave, with a constant fission rate during the in-core residence time and no fission during the ex-core residence time. This approximation simplifies the analytical form of the fission history for creating the non-linear least squares problem, as shown in Appendix A.

Figure 3.2 shows the flow diagram for a fission history to simulate irradiation on a fissile sample circulating in the primary loop. The model passes eight parameters into OpenMC [149]: in-core residence time, ex-core residence time, neutron irradiation energy, target fissile nuclide, target density, total irradiation time, temperature, and reprocessing terms. The external data passed into OpenMC includes the emission probabilities and the depletion matrix. The depletion matrix contains information on half-lives, fission yields, cross-sections, branching ratios, and other useful data needed for OpenMC depletion. Section 3.3 provides further details on this process.

After OpenMC completes the fixed-source simulation, the final nuclide concentrations in the irradiated sample are saved for analysis in MoSDeN. MoSDeN combines the concentrations with data from the IAEA beta-delayed neutron emission database to generate a delayed neutron count rate [49]. A non-linear least squares fitting function from SciPy then calculates the group half-lives and yields using the Trust Region Reflective algorithm [188, 26]. These group parameters correspond to a specific nuclide, neutron energy, in-core residence time, ex-core residence time, reprocessing scheme, and reprocessing scalar. Section 3.4 provides further details on this block. Some additional factors to consider include self-shielding due to sample density (while avoiding self-multiplication), temperature-induced cross-section variations, and under- or over-irradiation, all of which can affect the results [87].

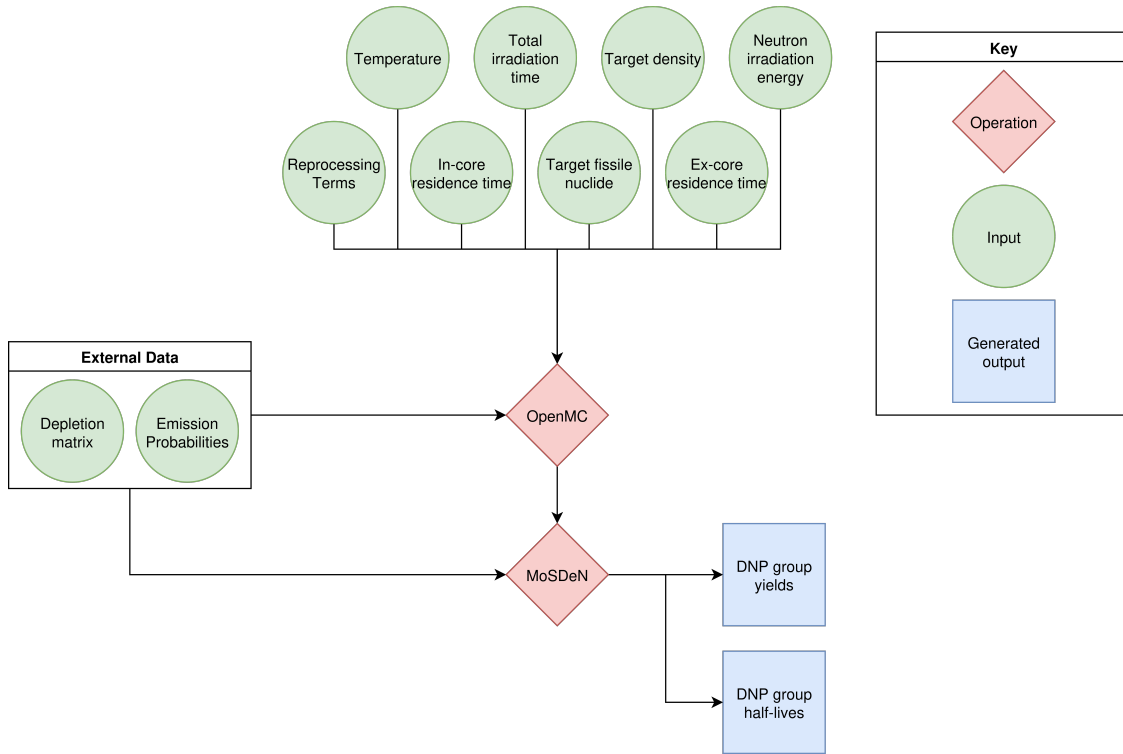


Figure 3.2: Flow diagram of 0D flow model used to generate improved DNP precursor group parameters.

The OpenMC model uses a three-gram spherical sample with vacuum boundary conditions. It treats the

problem as a fixed-source simulation, with a uniform neutron source distributed evenly throughout the sphere. This approach is not physical, as a real neutron source would have a spectrum and originate externally. However, the exact neutron spectrum for irradiating samples to generate DNP group parameters is not provided in the literature. The source distribution also has negligible impact, as the sample is small and exhibits minimal self-multiplication. Thus, this approximation does not limit the effectiveness of the 0D flow model.

Another consideration for the 0D flow model is maintaining a consistent source particle rate. While not strictly necessary for group parameter generation since the values are per-fission, it simplifies the data analysis, as the tallies depend on the source particles. Equation (3.7) provides the source rate for the irradiation simulations:

$$\dot{S} = S_{net} \frac{\tau_{in} + \tau_{ex}}{\tau_{in} T}, \quad (3.7)$$

where

- $\dot{S}$ = the source particle rate [$n \cdot s^{-1}$]
- S_{net} = the total number of source particles [n]
- τ_{in} = the in-core residence time [s]
- τ_{ex} = the ex-core residence time [s]
- T = the net irradiation time [s].

3.2.3 Spatially Resolved Model

The 0D flow model rapidly generates improved DNP group parameters, but it simplifies the treatment of flow. Instead of modeling the actual fuel salt flow, it estimates fuel circulation using residence times for each region. A spatially resolved model is useful for capturing the effects that the 0D flow model only capture with residence times for specific MSR geometries. Figure 3.3 shows the flow diagram for the spatially resolved model, which models the flow using Moltres.

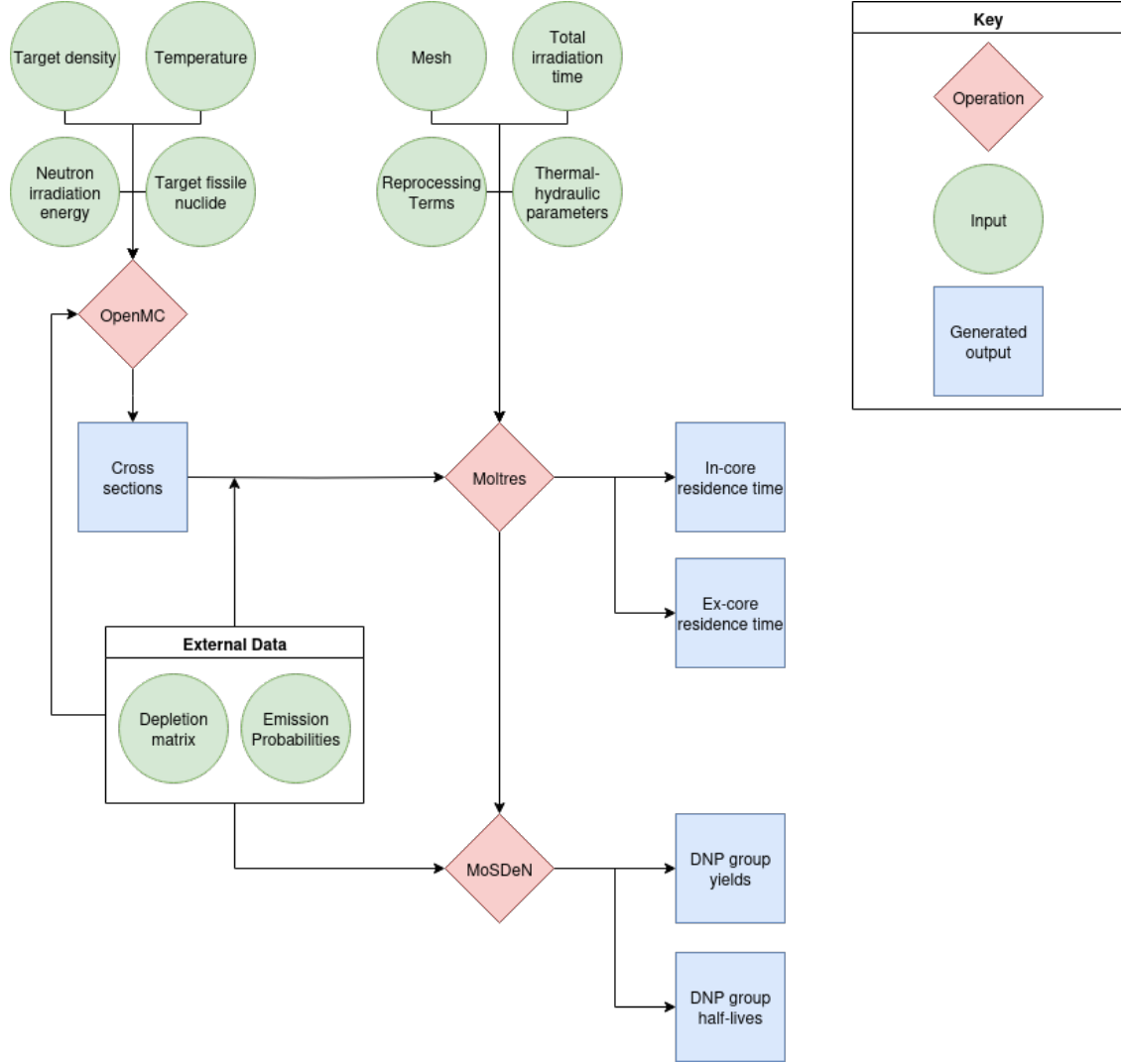


Figure 3.3: Flow diagram of spatially resolved flow model used to generate improved DNP group parameters.

The spatially resolved flow model introduces two new parameters absent in the 0D flow model. The first is the mesh, which defines the problem geometry. MOOSE provides built-in meshing tools, though complex geometries may require external tools such as Gmsh [65]. The second is the thermal-hydraulics parameters, which capture physical phenomena enabled by spatial resolution, as discussed in Section 3.1.3. Additionally, since the depletion happens in Moltres instead of OpenMC, OpenMC does not need as much data.

In the spatially resolved model, the in-core and ex-core residence times are outputs rather than inputs. These times derive from the movement of the irradiated sample over time in the fuel salt. The in-core and ex-core residence times vary as the irradiated sample diffuses and mixes further into the fuel salt. The net residence time, τ_{net} , remains constant unless the model changes, such as an increased temperature difference or induced flow rate. Equation (3.8) defines the in-core residence time as:

$$\tau_{in}(t) = \frac{\int_{V_{in}} \int_0^t N(\vec{x}, \tau) d\tau d\vec{x}}{\int_V \int_0^t N(\vec{x}, \tau) d\tau d\vec{x}} \tau_{net}, \quad (3.8)$$

where

$$\begin{aligned}
V_{in} &= \text{the in-core volume } [cm^3] \\
V &= \text{the total primary volume } [cm^3] \\
N(\vec{x}, t) &= \text{the time and space resolved sample concentration } [atoms \cdot cm^{-3}] \\
\tau_{net} &= \text{the net residence, or circulation, time of the fuel salt } [s].
\end{aligned}$$

The ex-core residence time can be calculated in the same way, or alternatively as shown in Equation (3.9) as:

$$\tau_{ex} = \tau_{net} - \tau_{in}. \quad (3.9)$$

3.3 Delayed Neutron Count Rate Generation

Although the various models have been discussed, execution details were not. The following sub-sections provide specific information on the methodology behind the generation of the improved DNP group parameters. This includes a detailed discussion on the concentration generation and the data used with those concentrations to produce a delayed neutron count rate.

3.3.1 DNP Concentrations

The concentrations from the 0D scaled model are calculated using Equation 3.6, while the concentrations in the spatially resolved model can be calculated using Equation 2.58. The 0D flow model will use OpenMC to handle calculation of the concentrations over time.

The 0D flow model will use a fixed-source problem on a three-gram spherical sample of a fissile target in OpenMC. A uniformly distributed source will be placed inside the target with a vacuum boundary condition at the edge of the sphere. The sample must have a sufficiently large volume to ensure a high number of fissions, but excessive volume increases self-shielding and self-multiplication. Adjusting the density mitigates this issue: higher density increases self-shielding and self-multiplication, while lower density reduces fission probability and worsens statistical accuracy. By using a uniformly distributed neutron source within the sample, these effects are lessened. Additionally, since the fission counts can be taken directly from OpenMC tallies, self-multiplication and self-shielding can be accounted for directly without introducing error.

The fixed-source in this model will consist entirely of neutrons at a single energy, rather than a spectrum. For the proposed analyses, only thermal fission of ^{235}U is studied. This energy dictates fission cross sections, transmutation cross sections, and fission yields. Traditionally, tabulated group parameter data is formatted for a single energy or energy range, such as fast or thermal. However, macroscopic experiments, discussed in Section 2.1.1, use reactors with varying neutron energy spectra, not mono-energetic neutrons. This could be addressed by modeling a reactor in OpenMC and using a k-eigenvalue solve to generate a neutron energy spectrum. This analysis is considered outside the current scope, but is suggested as future work in Section 5.4. Photons can also generate delayed neutrons through photofission, though this is also considered outside the current scope [90].

The neutron source distribution does not replicate the experimental process, where neutrons irradiate the sample externally rather than internally. However, an evenly distributed source eliminates self-shielding concerns. The exact spatial locations of fission events within the sample should negligibly affect the nuclide concentrations, which are the outputs of the OpenMC simulation. With a density of $10 \text{ g} \cdot \text{cm}^{-3}$, the spherical sample has a radius of

approximately 0.4 cm.

After running the OpenMC simulation, the concentrations are collected from the depletion results for the final irradiation time step. These final concentrations are then converted into a delayed neutron count rate.

3.3.2 Concentrations to Count Rate

After receiving the concentrations from OpenMC, MoSDeN determines the concentration of each nuclide as a function of time during the measurement period. The concentrations are calculated based on half-life, as shown in Equation (3.10), where the concentration as a function of time is:

$$N(t) = N_0 e^{-\lambda t}. \quad (3.10)$$

Equation (3.10) shows the approach used in the 0D scaled model, and does not account for decay chains between nuclides. However, the general form of the equation can still be used, but with inclusion of the various decay chains and branching ratios. A simple way of implementing decay with the decay chain is using zero-power depletion simulations in OpenMC. This is used in the 0D flow model, though the 0D flow model also offers the option of using Equation (3.10). These concentrations are available at coarse time steps and linearly interpolated using SciPy, allowing MoSDeN to access more accurate concentrations at any time after irradiation [186]. The specific time steps at which concentrations are written will be investigated after development of the 0D flow model.

The concentrations are combined with Equation (2.2), recreated here:

$$\dot{n}_d(t) = \epsilon \sum_{k=1}^K P_{n,k} \lambda_k N_k(t).$$

The efficiency term, ϵ , is set to 1 in this work, as the simulated method includes every produced delayed neutron in the count rate. The contributions of each DNP over time are summed to generate the delayed neutron count rate.

3.4 DNP Group Reconstruction

Once the delayed neutron count rate is generated, it is used to generate the improved DNP group parameters, $\bar{v}_{d,k}$ and λ_k . Fitting the DNP group parameters to the delayed neutron count rate curve requires a non-linear least squares method, as both the group half-lives and yields must be determined. The equation describing the problem appears in Equation (2.3), recreated here:

$$\dot{n}_d(t) \approx \epsilon F_s \sum_{k=1}^K \bar{v}_{d,k} \lambda_k e^{-\lambda_k t}.$$

An important consideration for this process is the selection of time values for generating the DNP group parameters. Increasing the number of time nodes raises computational costs, but too few nodes result in a poorer fit [126]. Section 4.2.2 shows the sensitivity study for the time nodes, for both the node spacing and the maximum time to measure delayed neutrons.

Another consideration for the non-linear least squares solve is the covariance, which is larger for the pulse irradiation than for the saturation irradiation. This results from multiplying the group yield by the group decay constant. Attempts to replace this multiplication with an independent term produces worse fits, as the multiplication is not independent and is necessary for accurate modeling.

The residual of the solve is given in Equation (2.10), recreated here as:

$$r = \min_x \left\| \frac{b - Ax}{b + \epsilon} \right\|_2^2.$$

Half-life bounds are set from 1 millisecond to 1000 seconds, and the yield bounds are from 0 to 1. Convergence tolerances for the cost function, change in independent variables, and gradient norm are all set to 1×10^{-12} to ensure tight convergence.

The final consideration in DNP group reconstruction is uncertainty. This work tracks uncertainties using the uncertainties package [55]. These uncertainties include uncertainties in fission yields, half-lives, and emission probabilities. The uncertainties are propagated through the count rate calculation, but for the non-linear least squares solve a Monte Carlo sampling based approach is used. Each count rate is generated using randomly sampled fission yields, half-lives, and emission probabilities within their uncertainty bounds for each DNP. This enables formation of a normal distribution of group parameters, from which the mean and standard deviation of the mean can be extracted.

Once the DNP group parameters and their uncertainties are generated, the final step to obtain the finalized group parameters is to combine the saturation and pulse irradiation results. Saturation irradiation provides better statistics for longer-lived groups, while pulse irradiation is better for shorter-lived groups. Brady combines the results by using four long-period groups from saturation irradiation and four short-period groups from pulse irradiation. The groups are renormalized at the second group, where the two longest-lived group half-lives and yield ratios use the saturation results [25]. The remaining groups use the pulse irradiation results.

This work uses a similar but slightly simplified approach. The group parameters are first calculated for both pulse and saturation irradiation. Then, the two longest-lived groups use the saturation irradiation results, while the four shortest-lived groups use the pulse irradiation results.

3.5 Impact of DNP Groups on Transients

The impact of the DNP groups on transients will be evaluated using Moltres with fluid-flow-velocity field coupled multi-group diffusion kinetics, as discussed in Section 2.1.3. This approach provides high-fidelity results by coupling thermal-hydraulics and neutronics, a necessity for MSR modeling. Modeling every DNP in space and time would be prohibitively expensive, so they are instead modeled using DNP group parameters. Moltres allows direct comparison between static DNP group parameters, which do not account for MSR phenomena, and parameters that do.

The results of interest from this analysis include the reactivity and power responses to various initiating events. The initiating events and transients vary, as different reactor responses are studied. The events of interest depend on the perspective from which the reactor response is analyzed.

Specifically, there are four different perspectives of interest in this work: validation, safety, security, and safeguards. Each perspective involves a unique combination of motivation, proposed initiating event, and investigated response. These are discussed in detail in their respective sections.

There are some general results of interest, such as the effective delayed neutron fraction. This can be generated from Moltres using the prompt method, discussed in Section 2.1.2. Moltres uses fission and adjoint-weighted DNP group parameters, and the effective multiplication factor will capture the spatial distribution of each DNP group concentration. This will provide an effective delayed neutron fraction that captures the effects of fuel circulation. The fission weighting and adjoint flux values will be generated using OpenMC. The fission weights can be calculated using fission rate tallies, while the adjoint flux can be calculated using a separate coarse mesh finite difference

(CMFD) OpenMC run, which enables the adjoint flux to be solved on a cartesian grid.

3.5.1 Validation

The motivation for validation is to confirm that the proposed methodology and results align with experimental data. This ensures the demonstrated improved DNP group parameters are applicable to actual MSRs. Discrepancies between experimental results and the model highlight potential weaknesses in the approach. If no discrepancies exist, confidence in the results increases for problems similar to those validated.

The validation of the DNP group parameters follows a multi-stage approach. The first stage compares the irradiated sample's delayed neutron count rate to the Keepin et al. experiment from 1957 [87]. If the count rates do not match, the issue lies in the data or the aggregation of individual DNPs into the count rate. If the count rate is accurately predicted, the formulation is considered valid. The second stage validates the improved DNP group parameters using MSRE pump start-up and coast-down experimental data to assess their accuracy in modeling reactivity changes from DNPs [139]. This experiment has been studied using Moltres and Quasimolto, allowing comparison with their results [142, 128]. However, previous models excluded the inlet and outlet plena, leading to discrepancies with experimental data. A model including the plena, combined with updated DNP group parameters, should yield results closer to the experimental data.

3.5.2 Safety

The motivation for studying safety is to protect the public and meet regulatory requirements. Safety is discussed in detail in Section 2.3.1, though the general takeaway is that many safety-related transients exist. The DNP group parameters are critical in these transients, as they influence the reactor period, power, and temperature.

The transients investigated are ULOHS, ULOF, UTOP, and FSOC. For each transient, the reactor period, power response, maximum temperature, minimum temperature, and reactivity response are recorded. The reactor period is critical, as it reflects the reactor's controllability. The power response shows how changes in the DNP group parameters affect other responses, such as temperature. It is important to make sure differences in the DNP parameters don't lead to temperature changes or oscillations which result in unsafe reactor conditions.

3.5.3 Security

Security is discussed in detail in Section 2.3.2, but the general takeaway is that bad actors may attempt to sabotage the reactor. This ties directly into the safety design study, as sabotage aims to create unsafe reactor conditions. Ensuring the MSR operates safely after sabotage is critical. If static DNP group parameters predict safe operation while improved parameters predict unsafe operation, correcting this discrepancy in the reactor design is essential.

Since the safety study covers most transients relevant to sabotage, additional considerations can be incorporated into these transients. As sabotage may initiate multiple safety-related transients simultaneously, it is reasonable to investigate such scenarios. With four unique transients, this results in six double-transient combinations, three triple-transient combinations, and one quadruple-transient combination, totaling ten security-related sabotage-initiated transients.

3.5.4 Safeguards

Safeguards are discussed in Section 3.5.4, but the main takeaway is the importance of preventing SNM diversion in MSRs. Detection methods, such as neutron detectors and surveillance systems, play a key role in safeguarding the

reactor [200]. The improved DNP group parameters in this work affect the effective delayed neutron fraction and reactivity response. Using static DNP group parameters to detect SNM diversion predicts less accurate reactivity responses than the improved parameters. If the difference is significant, relying on static parameters may reduce the detection probability of a diversion scenario.

To investigate this, diversion scenarios are studied for a single significant quantity of ^{235}U and Pu removed over the course of one year, simulated using OpenMC. The change in the fuel salt composition affects the DNP group parameters, as discussed in Section 2.1.2. The importance and fission-weighted DNP group parameters can be used in Moltres, as discussed in Section 3.1.3, for simple step reactivity insertion transients, where the following four sets of data are compared: static group parameters without diversion, static group parameters with diversion, improved group parameters without diversion, and improved group parameters with diversion. If the difference between the static group parameters with and without diversion is smaller or similar to the difference between the static and improved group parameters, then this means detection of that diversion scenario requires models to account for the MSR physical phenomena when generating DNP group parameters.